1 Gain law

(bars denote normalized quantities: lengths by R, $\bar{a} = a/a_o$, $\bar{f} = a_o f$; $\bar{w} > \bar{w}_s$, $b > 0$, $p > 0$; $e_\theta = \theta - \hat{\theta}$)

$$\begin{aligned}
\bar{a}(\bar{w} - \bar{w}_s,\, b,\, p) &= 1 - \Big(1 + \frac{\bar{w} - \bar{w}_s}{b}\Big)^{-p} \qquad (\bar{w}_s - 1:\ \text{unknown surface position})\\
\bar{a}(\bar{w} - \bar{w}_s,\, b_w,\, p_w) &= 1 - \Big(1 + \frac{\bar{w} - \bar{w}_s}{b_w}\Big)^{-p_w}\\
\bar{a}\big(\bar{w}_d - \hat{\bar{w}}_s,\, \hat{b}_w,\, \hat{p}_w\big) &= 1 - \Big(1 + \frac{\bar{w}_d - \hat{\bar{w}}_s}{\hat{b}_w}\Big)^{-\hat{p}_w}
\end{aligned}$$

$$a_w = a_o\, \bar{a}(\bar{w} - \bar{w}_s,\, b_w,\, p_w), \qquad a_o = \frac{\Delta t}{\gamma_N R} \quad (a_o \text{ fixed, not a state})$$

$$p = 1: \qquad \bar{a} = \frac{\bar{w} - \bar{w}_s}{(\bar{w} - \bar{w}_s) + b}, \qquad \frac{d\bar{a}}{d\bar{w}} = \frac{b}{\big((\bar{w} - \bar{w}_s) + b\big)^2}$$

2 Gain rate and parameter Jacobians

$$\begin{aligned}
\bar{a}'(\bar{w} - \bar{w}_s,\, b,\, p) := \frac{d\bar{a}}{d\bar{w}} &= \frac{p}{b}\Big(1 + \frac{\bar{w} - \bar{w}_s}{b}\Big)^{-p-1}\\
\bar{a}'\big(\bar{w}_d - \hat{\bar{w}}_s,\, \hat{b}_w,\, \hat{p}_w\big) &= \frac{\hat{p}_w}{\hat{b}_w}\Big(1 + \frac{\bar{w}_d - \hat{\bar{w}}_s}{\hat{b}_w}\Big)^{-\hat{p}_w - 1}
\end{aligned}$$

$$\frac{\partial \bar{a}}{\partial \bar{w}_s} = -\,\bar{a}' \qquad (\text{exact, any } b,\, p)$$

$$\begin{aligned}
&\bar{a}'(\bar{w}_d - \bar{w}_s,\, b_w,\, p_w) - \bar{a}'\big(\bar{w}_d - \hat{\bar{w}}_s,\, \hat{b}_w,\, \hat{p}_w\big)\\
&\qquad = J_b\big(\bar{w}_d - \hat{\bar{w}}_s, \hat{b}_w, \hat{p}_w\big)\, e_{b_w} + J_p\big(\bar{w}_d - \hat{\bar{w}}_s, \hat{b}_w, \hat{p}_w\big)\, e_{p_w} + J_{w_s}\big(\bar{w}_d - \hat{\bar{w}}_s, \hat{b}_w, \hat{p}_w\big)\, e_{\bar{w}_s}
\end{aligned}$$

$$\begin{aligned}
J_b\big(\bar{w}_d - \hat{\bar{w}}_s, \hat{b}_w, \hat{p}_w\big) &= \Big[-\frac{1}{\hat{b}_w} + \frac{(\hat{p}_w + 1)\,(\bar{w}_d - \hat{\bar{w}}_s)}{\hat{b}_w\big((\bar{w}_d - \hat{\bar{w}}_s) + \hat{b}_w\big)}\Big]\, \bar{a}'\big(\bar{w}_d - \hat{\bar{w}}_s, \hat{b}_w, \hat{p}_w\big)\\
J_p\big(\bar{w}_d - \hat{\bar{w}}_s, \hat{b}_w, \hat{p}_w\big) &= \Big[\frac{1}{\hat{p}_w} - \ln\Big(1 + \frac{\bar{w}_d - \hat{\bar{w}}_s}{\hat{b}_w}\Big)\Big]\, \bar{a}'\big(\bar{w}_d - \hat{\bar{w}}_s, \hat{b}_w, \hat{p}_w\big)\\
J_{w_s}\big(\bar{w}_d - \hat{\bar{w}}_s, \hat{b}_w, \hat{p}_w\big) &= \Big[\frac{\hat{p}_w + 1}{(\bar{w}_d - \hat{\bar{w}}_s) + \hat{b}_w}\Big]\, \bar{a}'\big(\bar{w}_d - \hat{\bar{w}}_s, \hat{b}_w, \hat{p}_w\big)
\end{aligned}$$

3 True dynamics

$\big(\delta\bar{w}_j[k] := \delta\bar{w}[k-d+j-1], \quad j = 1,\ldots,d{+}1, \quad d=2; \qquad \delta\bar{w}_D \equiv 0 \text{ hereafter}\big)$

$$
\begin{aligned}
\bar{w}[k] &= \bar{w}_d[k] - \delta\bar{w}_3[k] \\
\bar{a}_w[k+1] &= \bar{a}_w\big(\bar{w}[k+1]\big) = \bar{a}\big(\bar{w}_d[k+1] - \bar{w}_s - \delta\bar{w}_3[k+1],\, b_w,\, p_w\big) \\
\bar{w}_d[k+1] &= \bar{w}_d[k] + \Delta\bar{w}_d[k]
\end{aligned}
$$

$$
\begin{aligned}
\delta\bar{w}_1[k+1] &= \delta\bar{w}_2[k] \\
\delta\bar{w}_2[k+1] &= \delta\bar{w}_3[k] \\
\delta\bar{w}_3[k+1] &= \lambda_c\,\delta\bar{w}_3[k] - F_{dw}[k]\,e_{\bar{a}_w}[k] - \varepsilon_w[k] \qquad (e_{\bar{a}_w} = \bar{a}_w - \hat{\bar{a}}_w) \\
b_w[k+1] &= b_w[k] \\
p_w[k+1] &= p_w[k] \\
\bar{w}_s[k+1] &= \bar{w}_s[k]
\end{aligned}
$$

$$
\begin{aligned}
F_{dw}[k] &:= \bar{f}_{dw}[k] + (1-\lambda_c)\sum_{i=1}^{d}\bar{f}_{dw}[k-i] \\
\varepsilon_w[k] &:= \bar{a}_w[k]\bar{f}_T[k] + (1-\lambda_c)\sum_{i=1}^{d}\bar{a}_w[k-i]\bar{f}_T[k-i] + (1-\lambda_c)\,n_w[k]
\end{aligned}
$$

4 Process model

$$
\bar{w}[k+1] - \bar{w}[k] = \Delta\bar{w}_d[k] + (1-\lambda_c)\,\delta\bar{w}_3[k] + F_{dw}[k]\,e_{\bar{a}_w}[k] + \varepsilon_w[k]
$$

$$
\begin{aligned}
\bar{a}_w[k+1] &= \bar{a}_w[k] + \bar{a}'[k]\Big[\Delta\bar{w}_d[k] + (1-\lambda_c)\,\delta\bar{w}_3[k] + F_{dw}[k]\,e_{\bar{a}_w}[k] + \varepsilon_w[k]\Big] \\
\bar{a}'[k] &= \bar{a}'\big(\bar{w}_d[k] - \bar{w}_s[k],\, b_w[k],\, p_w[k]\big)
\end{aligned}
$$

$$
\mathbf{x}[k] = \big[\,\delta\bar{w}_1,\ \delta\bar{w}_2,\ \delta\bar{w}_3,\ \bar{a}_w,\ b_w,\ p_w,\ \bar{w}_s\,\big]^\top[k] \qquad (d=2)
$$

$$
\begin{aligned}
\delta\bar{w}_1[k+1] &= \delta\bar{w}_2[k] \\
\delta\bar{w}_2[k+1] &= \delta\bar{w}_3[k] \\
\delta\bar{w}_3[k+1] &= \lambda_c\,\delta\bar{w}_3[k] - F_{dw}[k]\,e_{\bar{a}_w}[k] - \varepsilon_w[k] \\
\bar{a}_w[k+1] &= \bar{a}_w[k] + \bar{a}'[k]\Big[\Delta\bar{w}_d[k] + (1-\lambda_c)\,\delta\bar{w}_3[k] + F_{dw}[k]\,e_{\bar{a}_w}[k] + \varepsilon_w[k]\Big] \\
b_w[k+1] &= b_w[k] \\
p_w[k+1] &= p_w[k] \\
\bar{w}_s[k+1] &= \bar{w}_s[k]
\end{aligned}
$$

5 Estimator

$$
\begin{aligned}
\delta\hat{\bar{w}}_1[k+1] &= \delta\hat{\bar{w}}_2[k] \\
\delta\hat{\bar{w}}_2[k+1] &= \delta\hat{\bar{w}}_3[k] \\
\delta\hat{\bar{w}}_3[k+1] &= \lambda_c\,\delta\hat{\bar{w}}_3[k] \\
\hat{\bar{a}}_w[k+1] &= \hat{\bar{a}}_w[k] + \hat{\bar{a}}'[k]\,\Big[\Delta\bar{w}_d[k] + (1-\lambda_c)\,\delta\hat{\bar{w}}_3[k]\Big] \\
\hat{b}_w[k+1] &= \hat{b}_w[k] \\
\hat{p}_w[k+1] &= \hat{p}_w[k] \\
\hat{\bar{w}}_s[k+1] &= \hat{\bar{w}}_s[k]
\end{aligned}
$$

$$\hat{\bar{a}}'[k] = \bar{a}'\big(\bar{w}_d[k] - \hat{\bar{w}}_s[k],\, \hat{b}_w[k],\, \hat{p}_w[k]\big)$$

6 Error dynamics

(True − Estimator; coefficients at the estimate)

$$e_{\delta\bar{w}_j} = \delta\bar{w}_j - \delta\hat{\bar{w}}_j \quad (j = 1, 2, 3), \qquad \mathbf{e} = \big[\, e_{\delta\bar{w}_1},\ e_{\delta\bar{w}_2},\ e_{\delta\bar{w}_3},\ e_{\bar{a}_w},\ e_{b_w},\ e_{p_w},\ e_{\bar{w}_s} \,\big]^\top$$

$$
\begin{aligned}
e_{\delta\bar{w}_1}[k+1] &= e_{\delta\bar{w}_2}[k] \\
e_{\delta\bar{w}_2}[k+1] &= e_{\delta\bar{w}_3}[k] \\
e_{\delta\bar{w}_3}[k+1] &= \lambda_c\, e_{\delta\bar{w}_3}[k] - F_{dw}[k]\, e_{\bar{a}_w}[k] - \varepsilon_w[k] \\
e_{\bar{a}_w}[k+1] &= \big[1 + \hat{\bar{a}}'[k]\, F_{dw}[k]\big]\, e_{\bar{a}_w}[k] + J_{b_w}[k]\big[\Delta\bar{w}_d[k] + (1-\lambda_c)\,\delta\hat{\bar{w}}_3[k]\big]\, e_{b_w}[k] \\
&\quad + J_{p_w}[k]\big[\Delta\bar{w}_d[k] + (1-\lambda_c)\,\delta\hat{\bar{w}}_3[k]\big]\, e_{p_w}[k] + J_{w_s}[k]\big[\Delta\bar{w}_d[k] + (1-\lambda_c)\,\delta\hat{\bar{w}}_3[k]\big]\, e_{\bar{w}_s}[k] \\
&\quad + (1-\lambda_c)\,\hat{\bar{a}}'[k]\, e_{\delta\bar{w}_3}[k] + \hat{\bar{a}}'[k]\,\varepsilon_w[k] \\
e_{b_w}[k+1] &= e_{b_w}[k] \\
e_{p_w}[k+1] &= e_{p_w}[k] \\
e_{\bar{w}_s}[k+1] &= e_{\bar{w}_s}[k]
\end{aligned}
$$

$$
\begin{aligned}
J_{b_w}[k] &= J_b\big(\bar{w}_d[k] - \hat{\bar{w}}_s[k],\, \hat{b}_w[k],\, \hat{p}_w[k]\big) \\
J_{p_w}[k] &= J_p\big(\bar{w}_d[k] - \hat{\bar{w}}_s[k],\, \hat{b}_w[k],\, \hat{p}_w[k]\big) \\
J_{w_s}[k] &= J_{w_s}\big(\bar{w}_d[k] - \hat{\bar{w}}_s[k],\, \hat{b}_w[k],\, \hat{p}_w[k]\big)
\end{aligned}
$$

$$\mathbf{e}[k+1] = \mathbf{F}_e[k]\,\mathbf{e}[k] + \mathbf{q}[k]$$

$$\mathbf{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dw} & 0 & 0 & 0 \\ 0 & 0 & (1-\lambda_c)\hat{\bar{a}}' & 1+\hat{\bar{a}}' F_{dw} & J_{b_w}\big[\Delta\bar{w}_d+(1-\lambda_c)\delta\hat{\bar{w}}_3\big] & J_{p_w}\big[\Delta\bar{w}_d+(1-\lambda_c)\delta\hat{\bar{w}}_3\big] & J_{w_s}\big[\Delta\bar{w}_d+(1-\lambda_c)\delta\hat{\bar{w}}_3\big] \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{q}[k] = \big[0,\ 0,\ -\varepsilon_w[k],\ \hat{\bar{a}}'[k]\,\varepsilon_w[k],\ 0,\ 0,\ 0\big]^\top \qquad \big(q_4 = -\hat{\bar{a}}'[k]\,q_3\big)$$

7 Measurement and innovation

Measurement (d-step delay; $\bar{a}_{wm} = a_{wm}/a_o$; y_2 raw readout — AR(1) whitening enters with R_2, S8)

$$\begin{aligned} \nabla_d\bar{w}_d[k] &:= \bar{w}_d[k] - \bar{w}_d[k-d] \\ y_1[k] &= \delta\bar{w}_m[k] = \delta\bar{w}_1[k] + n_w[k] \\ y_2[k] &= \bar{a}_{wm}[k] = \bar{a}_w[k-d] + n_a[k] \end{aligned}$$

Delay back-off

$$\bar{a}_w[k-d] = \bar{a}_w[k] - \bar{a}'[k]\,\nabla_d\bar{w}_d[k] \qquad \Longrightarrow \qquad y_2[k] = \bar{a}_w[k] - \bar{a}'[k]\,\nabla_d\bar{w}_d[k] + n_a[k]$$

Linearization (partials at the estimate; $\bar{a}'$ does not depend on the state $\bar{a}_w$, so $\partial y_2/\partial\bar{a}_w$ is exactly 1)

$$\frac{\partial y_2}{\partial \bar{a}_w} = 1, \qquad \frac{\partial y_2}{\partial b_w} = -\nabla_d\bar{w}_d[k]\,J_{b_w}[k], \qquad \frac{\partial y_2}{\partial p_w} = -\nabla_d\bar{w}_d[k]\,J_{p_w}[k], \qquad \frac{\partial y_2}{\partial \bar{w}_s} = -\nabla_d\bar{w}_d[k]\,J_{w_s}[k]$$

Output equation ($\mathbf{H}$ is the Jacobian; row 2 is nonlinear in $\mathbf{x}$, so $\mathbf{Hx}$ is its linearization, not an identity)

$$\mathbf{y}[k] = \mathbf{H}[k]\,\mathbf{x}[k] + \mathbf{r}[k], \qquad \mathbf{r}[k] = \begin{bmatrix} n_w[k] \\ n_a[k] \end{bmatrix}$$

$$\mathbf{H}[k] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & -\nabla_d\bar{w}_d\,J_{b_w} & -\nabla_d\bar{w}_d\,J_{p_w} & -\nabla_d\bar{w}_d\,J_{w_s} \end{bmatrix}$$

Innovation

$$
\begin{aligned}
e_{y_1}[k] &= y_1[k] - \delta\hat{\bar{w}}_1[k] = e_{\delta\bar{w}_1}[k] + n_w[k] \\
e_{y_2}[k] &= y_2[k] - \hat{\bar{a}}_w[k] + \hat{\bar{a}}'[k]\,\nabla_d\bar{w}_d[k] \\
&= e_{\bar{a}_w}[k] - \nabla_d\bar{w}_d[k]\,\big[J_{b_w}[k]\,e_{b_w}[k] + J_{p_w}[k]\,e_{p_w}[k] + J_{w_s}[k]\,e_{\bar{w}_s}[k]\big] + n_a[k]
\end{aligned}
$$

8 Process and measurement noise

Process noise Q (full $\mathrm{Var}(\varepsilon_w)$ container — thermal history and n_w feedthrough included, only the cross-step correlations of ε_w dropped; revision 2026-08-01, supersedes the current-step-only container; see ref)

$$
\begin{aligned}
&q_3[k] = -\bar{a}_w[k]\,\bar{f}_T[k], \qquad q_4[k] = \hat{\bar{a}}'[k]\,\bar{a}_w[k]\,\bar{f}_T[k] = -\hat{\bar{a}}'[k]\,q_3[k] \\
&\gamma(\bar{w}) = \gamma_N\,c(\bar{w}), \qquad \sigma^2_{\bar{f}_T} = a_o^2\,\frac{4k_BT\,\gamma(\bar{w})}{\Delta t}, \qquad \bar{a}_w = \frac{\Delta t}{a_o\,\gamma(\bar{w})\,R} \qquad \Longrightarrow \qquad \mathrm{Var}\big(\bar{a}_w\bar{f}_T\big) = \frac{4k_BT\,a_o}{R}\,\bar{a}_w
\end{aligned}
$$

$$
\begin{aligned}
&Q_{33}[k] = \mathrm{Var}(\varepsilon_w[k]) = \frac{4k_BT\,a_o}{R}\left(\bar{a}_w[k] + (1-\lambda_c)^2\sum_{i=1}^{d}\bar{a}_w[k-i]\right) + (1-\lambda_c)^2\,\sigma^2_{n_w} \\
&Q_{34}[k] = Q_{43}[k] = -\hat{\bar{a}}'[k]\,Q_{33}[k], \qquad Q_{44}[k] = \hat{\bar{a}}'[k]^2\,Q_{33}[k] \\
&Q_{55} = Q_{66} = Q_{77} = 0 \quad (b_w,\ p_w,\ \bar{w}_s \text{ constant})
\end{aligned}
$$

All other entries vanish; the gain block is rank one, since $q_4 = -\hat{\bar{a}}'[k]\,q_3$. (run time: entries at the current estimate)

Measurement noise $R = \mathrm{diag}(R_1, R_2)$ (a_{cov} = EWMA coefficient, IF_{var} = MA(d) inflation factor of $\hat{\sigma}^2_{\delta\bar{w}_r}$; y_2 treated as white here — whitened treatment in the ref notes)

$$
R_1 = \sigma^2_{n_w}
$$

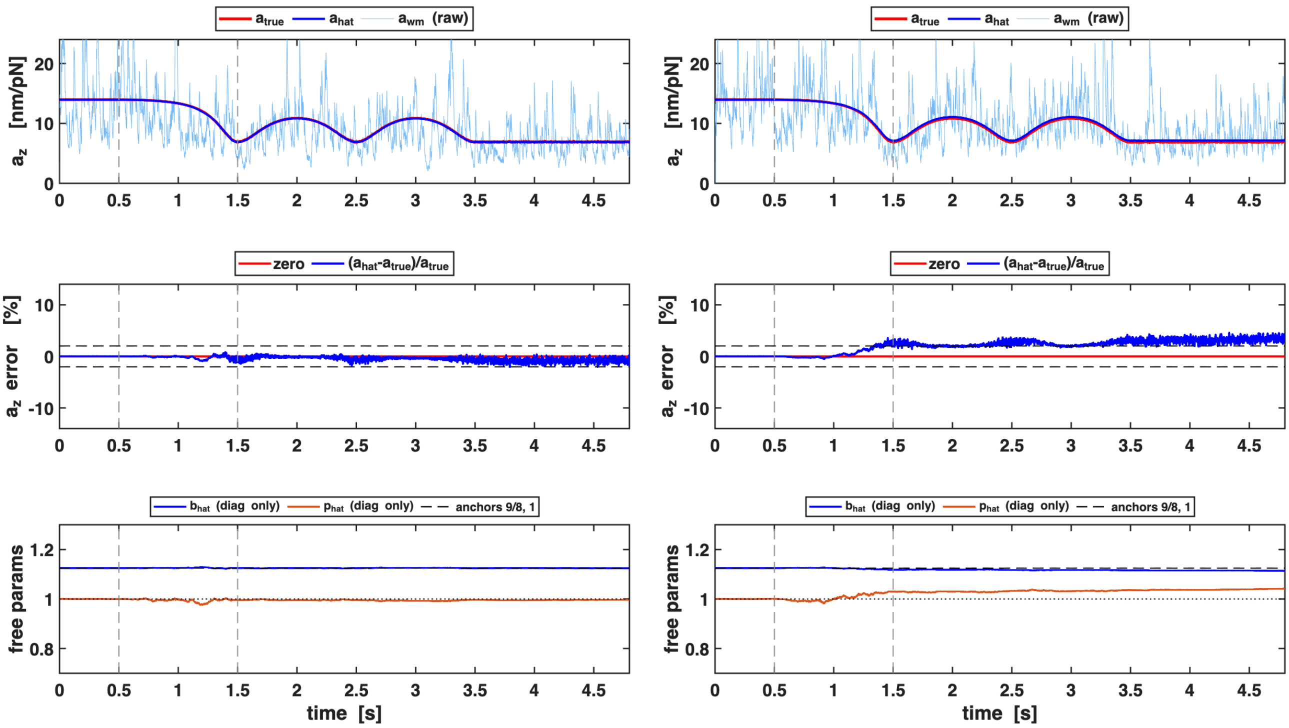
$$
\begin{aligned}
\sigma^2_{\delta\bar{w}_r}[k] &= C_{dpmr}\,\frac{4k_BT\,a_o}{R}\,\bar{a}_w[k] + C_n\,\sigma^2_{n_w} = C_{dpmr}\,\frac{4k_BT\,a_o}{R}\,\big(\bar{a}_w[k] + \xi\big) \\
\xi &:= \frac{C_n}{C_{dpmr}}\,\frac{\sigma^2_{n_w}}{4k_BT\,a_o/R} \\
\bar{a}_{wm}[k] &= \frac{\hat{\sigma}^2_{\delta\bar{w}_r}[k] - C_n\,\sigma^2_{n_w}}{C_{dpmr}\,\big(4k_BT\,a_o/R\big)} \\
\mathrm{Var}\big(\hat{\sigma}^2_{\delta\bar{w}_r}\big) &= K_{var}\,IF_{var}\,\big(\sigma^2_{\delta\bar{w}_r}\big)^2, \qquad K_{var} = \frac{2a_{cov}}{2-a_{cov}}
\end{aligned}
$$

$$
R_2[k] = K_{var}\,IF_{var}\,\big(\bar{a}_w[k] + \xi\big)^2 + d\,Q_{44}[k]
$$

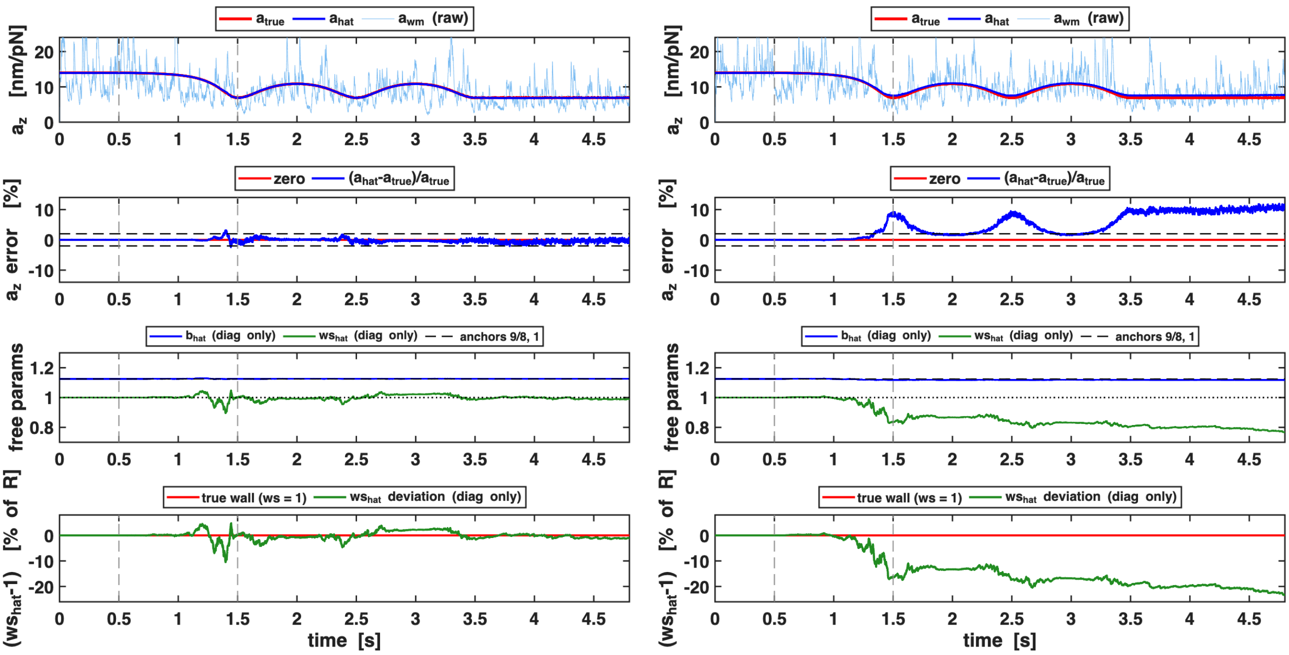
(C_{dpmr}, C_n: `Cdpmr_Cn_derivation.tex` — stationary, noise-only derivation; the former "defect 2" motion excess was RETRACTED 2026-08-01 (N=48: +1.33% n.s.); the live open item is the per-seed finite-window readout fluctuation, std 5–6%; see ref)

Simulation figures

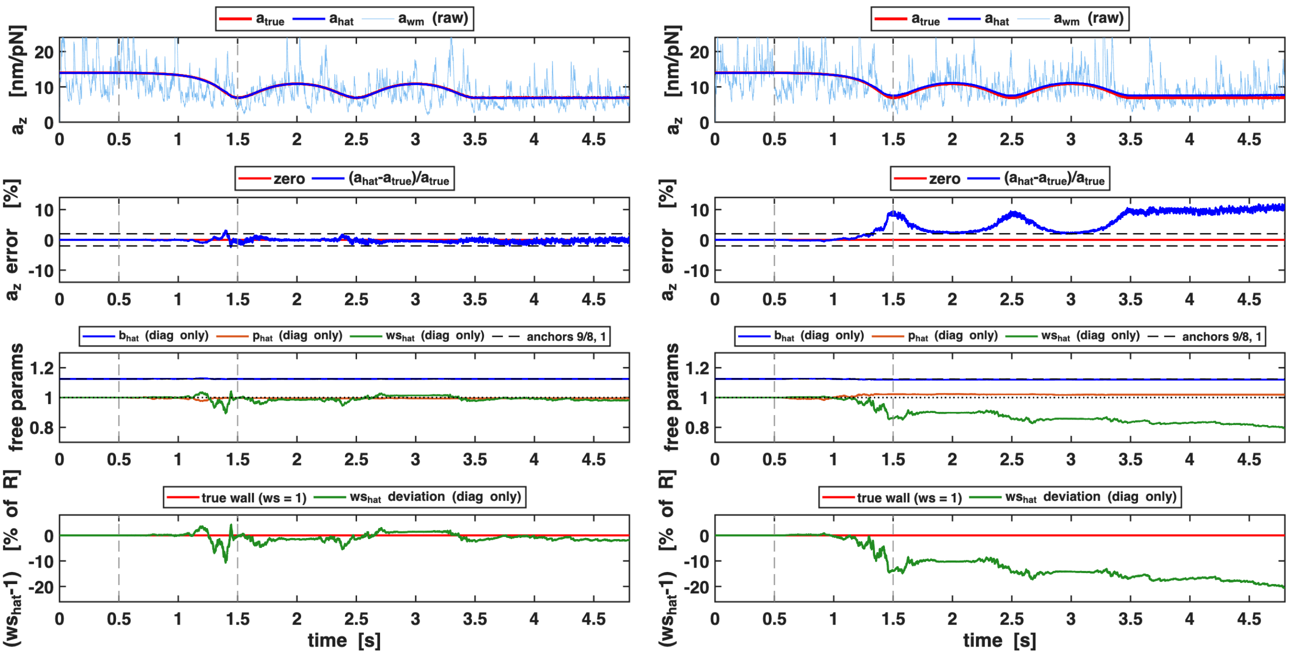
(canonical scenario, z axis; per panel: gain tracking, relative gain error, free parameters)



Config 1 — estimate (b, p), w_s locked at 1: seed 7 (left) vs seed 11 (right).



Config 2 — p locked at 1, estimate (b, w_s): seed 7 vs seed 11.



Config 3 — estimate (b, p, w_s): seed 7 vs seed 11.